Md Masum Kawsar

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RESEARCH INTERESTS

Smart IoT Systems • Edge AI & TinyML • IoT Security • Embedded Machine Learning • Sensor Systems

EDUCATION

BSc in Computer Science & Engineering | Daffodil International University, Bangladesh **Sep 2021 – Jan 2026**
CGPA: 3.48/4.00

Higher Secondary Certificate (Science) | Collectorate School & College, Rangpur, Bangladesh **Passed: 2020**
GPA: 5.00/5.00

Secondary School Certificate (Science) | Raipur BL High School, Pirganj, Rangpur, Bangladesh **Passed: 2018**
GPA: 5.00/5.00

TECHNICAL SKILLS

IoT & Embedded Systems: ESP32 • Arduino • Embedded C/C++ • Circuit Design & Hardware Prototyping • Sensor Interfacing & Calibration (MOS Gas, Ultrasonic, Temperature/Humidity) • Blynk IoT Platform • OLED Displays • Actuator Control • Humidity Compensation • TinyML / On-Device Inference • Model-to-C Export • Environmental Sensing • IoT Intrusion Detection.

Machine Learning & AI: Machine Learning • Deep Learning • Transfer Learning • Explainable AI (SHAP, Grad-CAM) • Time-Series Forecasting • Gradient Boosting • ANN • SVM • Nested Grouped Cross-Validation • Hyperparameter Tuning • Statistical Testing (Wilcoxon, Holm Correction) • Model Evaluation (MCC, ROC-AUC, Macro-F1).

Application Areas: Edge AI • Medical Image Analysis • Computer Vision • Audio Classification.

Data Handling & Processing: NumPy • Pandas • Feature Extraction (MFCC, HOG, LBP, Color Histograms) • Data Augmentation • Image, Audio & Sensor Data Preprocessing • Statistical Analysis.

Frameworks & Libraries: TensorFlow • Keras • Scikit-learn • XGBoost • LightGBM • SHAP • OpenCV • Scikit-image • Librosa • Matplotlib • Flask.

Programming Languages: Python • C • C++ • Java.

Development Tools: Git • GitHub • Google Colab • Jupyter Notebook • VS Code • Cursor AI.

Relevant Coursework: Big Data and IoT • Microprocessor, Embedded System and IoT • Computer Networks • Information Security • Artificial Intelligence • Data Mining and Machine Learning.

RESEARCH EXPERIENCE

Low-Cost Indoor Air Quality Monitoring and Short-Term Forecasting on ESP32 Using a Humidity-Compensated MOS Sensor and TinyML

Research Project

- Designed and hand-wired the complete hardware circuit of a low-cost indoor air quality monitor using an ESP32, a Winsen MP-135 MOS gas sensor, and a DHT22 sensor; collected 46,897 readings over 17 days in an occupied room in Dhaka.
- Designed a humidity- and temperature-compensated, drift-robust relative Pollution Index using a rolling 3-hour baseline.
- Deployed a 60-tree LightGBM model as ~60 KB of C code on the ESP32 (86 µs per forecast), reducing forecast error by 13–16% versus a persistence baseline at 2-, 5-, and 10-minute horizons.
- Achieved class accuracies of 93.7%, 87.9%, and 82.4%; released the open dataset (doi.org/10.5281/zenodo.23888099), firmware, and a live web dashboard.

Local Labels Beat Public Data: Cross-Network Evaluation of Lightweight and Explainable IoT Intrusion Detectors

Research Project

- Evaluated five model families and nine microcontroller-sized models across four public NetFlow-v2 datasets over 10 seeds with Holm-corrected Wilcoxon tests.
- Showed that detectors reaching 0.94 MCC within a dataset fall to about zero on other networks (62% of cross-dataset pairs below 0.5 ROC-AUC), and explained the failure using SHAP and distribution-shift analysis.
- With 1,000 local labels, a 405-byte depth-4 decision tree reached 0.79 MCC, outperforming tuned Random Forest and XGBoost (≤0.76), and classified a flow in 2.4 µs on an ESP32.
- Released code and ESP32 benchmark: github.com/masumkawsar/iot-ids-cross-network

Comparative Evaluation of Machine Learning and Transfer-Learning-Based Deep Learning Models for Explainable Glaucoma Detection from Retinal Fundus Images

Research Project

- Compared SVM, XGBoost, DenseNet121, and EfficientNetB3 using approximately 6,000 retinal fundus images.
- Applied HOG, LBP, and color histogram features, transfer learning, and Grad-CAM for model explainability.
- Achieved 90.33% accuracy, 90.88% precision, 89.67% recall, 90.27% F1-score, and 97.13 AUC with DenseNet121.
- Deployed the model in GlaucomaDetect, a Flask-based web application.

Comparative Study of Optimized ANN and SVM Models for Environmental Sound Classification with Practical Web-Based Deployment

Research Project

- Compared optimized Artificial Neural Network (ANN) and Support Vector Machine (SVM) models using UrbanSound8K.
- Extracted 80-dimensional acoustic features from 40 MFCC coefficients and their means and standard deviations.
- Achieved 94.81% test accuracy with ANN and 94.66% with SVM using KerasTuner and GridSearchCV.
- Developed UrbanSoundAI, a Flask-based application featuring waveform, Mel spectrogram, and MFCC visualizations.

ACADEMIC PROJECTS

Smart Water Level Monitoring and Automatic Pump Control System

- Built an IoT-based water level monitor using an ESP32 and an ultrasonic sensor, displaying the live level on an OLED screen.
- Automated pump on/off control based on the water level, with a buzzer alert and remote monitoring through the Blynk IoT platform.

Obstacle-Avoiding Robot Car

- Built an autonomous robot car using an Arduino, an ultrasonic sensor, and a motor driver module that detects obstacles and steers around them without human control.

LIST OF PUBLICATIONS

- Mridu, Mayishat Altab., **Kawsar, Md Masum.** "Comparative Study of Optimized ANN and SVM Models for Environmental Sound Classification with Practical Web-Based Deployment," *Indian Journal of Computer Science and Technology (INDJCST)*, Vol. 5, Issue 2, pp. 904–910, May–August 2026. Published Online: July 23, 2026. DOI: https://doi.org/10.59256/indjcst.20260502098
- **Kawsar, Md Masum.** "Low-Cost Indoor Air Quality Monitoring and Short-Term Forecasting on ESP32 Using a Humidity-Compensated MOS Sensor and TinyML." Submitted to *SN Computer Science*.
- **Kawsar, Md Masum**, Mridu, Mayishat Altab. "Local Labels Beat Public Data: Cross-Network Evaluation of Lightweight and Explainable IoT Intrusion Detectors." Submitted to *SN Computer Science*.
- Mridu, Mayishat Altab., **Kawsar, Md Masum.** "Comparative Evaluation of Machine Learning and Transfer-Learning-Based Deep Learning Models for Explainable Glaucoma Detection from Retinal Fundus Images." Submitted to *Network Modeling Analysis in Health Informatics and Bioinformatics (NetMAHIB)*.